Generalized Anomaly Detection in Multi-Brand Industrial Control Systems Using Machine Learning

Capstone Project by- Sahil

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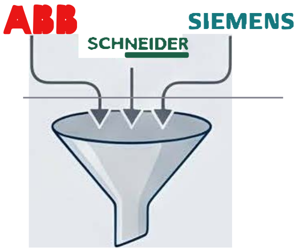
Objective:
developing an intelligent attack detection system using machine learning and deep learning techniques.

Attack Type

- Port scanning
- Nmap reconnaissance
- Smurf attack
- Ping of Death
- Teardrop attack

PHASE 1: DATA INGESTION & STRUCTURAL INTEGRITY



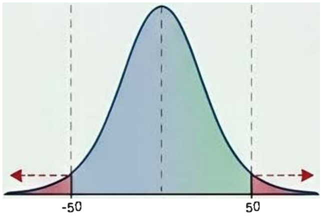
Multi-Vendor Ingestion & Tagging

Seamless CSV ingestion; automatic vendor tags for full traceability.

Schema Validation & Normalization

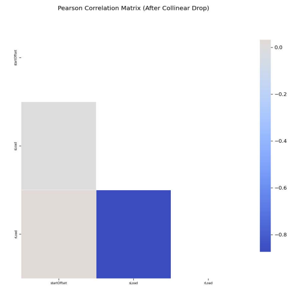
Automatically strips whitespace, validates labels, unilies fragmented attack names (e.g., 'IP, Scan' vs 'Ip scan') to canonical format.

PHASE 2: FEATURE REFINEMENT & DIMENSIONALITY REDUCTION



Winsorization & Outlier Management

IQR factor of 3.0 clamps extreme values. 166,040 values were clamped in one run to prevent model skew.



Collinear & Low-Variance Filtering

Drops features with Pearson correlation |r| > 0.05 and removes near-zero-variance columns.

One-Hot Encoding	Label Encoding
Vendor Typer Types	Protocols, Address...
	Protocols, Addresses

Memory-Safe Categorical Encoding

Implements One-Hot and Label Encoding to optimize memory usage.

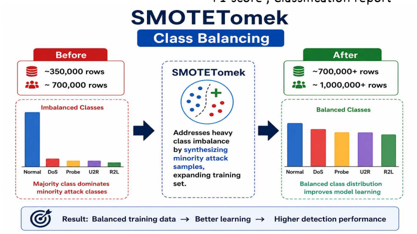
PHASE 4: CLASS BALANCING & MODEL PREPARATION

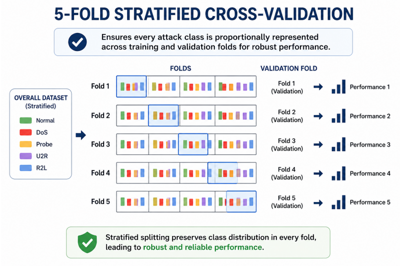
SMOTETomek

Class Balancing

Addresses heavy class imbalance by synthesizing minority attack samples, expanding training set

SMOTE was used to balance training data by generating synthetic minority-class examples. The evaluation code loaded trained models from disk and evaluated them on the test set.
For each model and task, it calculated: Accuracy, Precision, Recall F1-score , Classification report



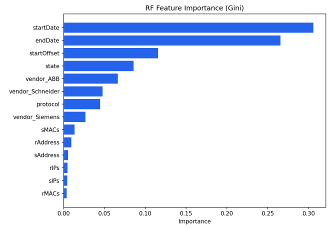


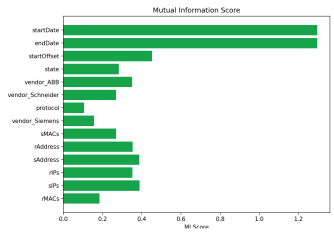
5-Fold Stratified Cross-Validation

Ensures every attack class is proportionally represented across training and validation folds for robust performance.

PHASE 3: CONSENSUS-BASED FEATURE SELECTION

THE TOP 15 FEATURESET





Final features selected via consensus of Random Forest Gini importance, Mutual Information, and Recursive Feature Elimination (RFE).

VIF Multicollinearity Check

A Variance Inflation Factor (VIF) audit detects multivariote correlations that simple penwise checks might miss.

PHASE 5: MODEL PERFORMANCE SHOWCASE

99.97%

F1-Score with Random Forest

Random Forest model achieved near-perfect accuracy and precision in multi-class attack detection tacks.

MODEL EVALUATION RESULTS (MULTI-CLASS)

Model	Accuracy	Precision	Recall	F1-Score
Random Forest (RF)	99.97%	99.97%	99.97%	0.9957
Decision Tree (DF)	99.97%	99.97%	99.97%	0.9997
KNN	99.87%	99.86%	99.87%	0.9986
Hybrid (DF-ANN)	99.14%	99.87%	99.14%	0.9949
Shallow ANN	99.00%	99.05%	99.08%	0.9942
Naive Bayes (HR)	65.40%	43.03%	65.40%	0.5106

Hybrid & Neural Network Flexibility

Pipeline supports diverse architectures including Shallow ANNs, Logistic Regression, and a Hybrid RF-ANN pipeline.